

What is an API?

An application programming interface is a way for two or more computer programs to communicate with each other. It is a type of software interface, offering a service to other pieces of software. A document or standard that describes how to build or use such a connection or interface is called an API specification.

The OpenWeatherMap API

<https://openweathermap.org/api>

The OpenWeatherMap is a service that provides weather data, including current weather data, forecasts, and historical data to the developers of web services and mobile applications.

It provides an API with JSON, XML, and HTML endpoints and a limited free usage tier. Making more than 60 calls per minute requires a paid subscription starting at USD 40 per month. Access to historical data requires a subscription starting at 150 USD per month. Users can request current weather information, extended forecasts, and graphical maps (showing cloud cover, wind speed, pressure, and precipitation).

To use this current weather data API, one must need the API key.

Pricing and API Key

<https://openweathermap.org/price>

API Error Code Meaning

<https://openweathermap.org/faq#error401>

Geocode API

<https://openweathermap.org/api/geocoding-api>

Used to obtain coordinates (latitude and longitude) of any city

```
In [ ]: import requests, json
```

```
In [ ]: def get_coordinates():  
        api_key = input('Enter your API key: ')  
        city = input('Enter the city name: ')  
        request_url = "http://api.openweathermap.org/geo/1.0/direct?q=" + city + "&appid=" + api_key  
        response = requests.get(request_url)
```

```

coordinates = response.json()
print(coordinates)
latitude = coordinates[0]['lat']
longitude = coordinates[0]['lon']
return (latitude, longitude)

```

```
lat, long = get_coordinates()
```

Enter your API key: 172dca3d92b0f6a90267417165f077dc

Enter the city name: Ahmedabad

```

[{'name': 'Ahmedabad', 'local_names': {'mr': 'अहमदाबाद', 'oc': 'Ahmadabad', 'ja': 'アフマダーバード', 'or': 'ଅହମଦାବାଦ', 'cs': 'Ahmadábád', 'feature_name': 'Ahmedabad', 'gu': 'અમદાવાદ', 'eo': 'Ahmadabado', 'ar': 'أحمد آباد', 'hi': 'अहमदाबाद', 'uk': 'Ахмедабад', 'pa': 'ਅਹਿਮਦਾਬਾਦ', 'ascii': 'Ahmedabad', 'de': 'Ahmedabad', 'ml': 'അഹ്മദാബാദ്', 'kn': 'ಅಹ್ಮದಾಬಾದ್', 'ur': 'احمد آباد', 'en': 'Ahmedabad', 'zh': '艾哈迈达巴德', 'ru': 'Ахмадабад', 'ta': 'அகமதாபாத்', 'he': 'אֲחַמְדַּאבָּאד', 'pl': 'Ahmadabad', 'te': 'అహ్మదాబాద్'}, 'lat': 23.0216238, 'lon': 72.5797068, 'country': 'IN', 'state': 'Gujarat'}]

```

```
In [ ]: print(lat)
```

```
23.0216238
```

```
In [ ]: print(long)
```

```
72.5797068
```

Current Weather API

<https://openweathermap.org/current>

Used to obtain current weather details of any city

```

In [ ]: def display_current_weather():
        api_key = input('Enter your API key: ')
        request_url = "https://api.openweathermap.org/data/2.5/weather?lat=" + str(lat)
        response = requests.get(request_url)
        current_weather = response.json()
        print(json.dumps(current_weather, indent = 5))
        description = current_weather['weather'][0]['description']
        temperature = current_weather['main']['temp']
        pressure = current_weather['main']['pressure']
        humidity = current_weather['main']['humidity']
        wind_speed = current_weather['wind']['speed']
        visibility = current_weather['visibility']
        print('Current Weather')
        print('Temperature', temperature)
        print('Pressure', pressure)
        print('Humidity', humidity)
        print('Wind Speed', wind_speed)
        print('Visibility', visibility)

display_current_weather()

```

Enter your API key: e9964ca2d303da57ca7c4b80c4afedf3

```
{
  "coord": {
    "lon": 72.5797,
    "lat": 23.0216
  },
  "weather": [
    {
      "id": 800,
      "main": "Clear",
      "description": "clear sky",
      "icon": "01d"
    }
  ],
  "base": "stations",
  "main": {
    "temp": 26.04,
    "feels_like": 26.04,
    "temp_min": 26.04,
    "temp_max": 26.04,
    "pressure": 1019,
    "humidity": 18
  },
  "visibility": 6000,
  "wind": {
    "speed": 3.09,
    "deg": 70
  },
  "clouds": {
    "all": 0
  },
  "dt": 1676272219,
  "sys": {
    "type": 1,
    "id": 9049,
    "country": "IN",
    "sunrise": 1676252625,
    "sunset": 1676293456
  },
  "timezone": 19800,
  "id": 1279233,
  "name": "Ahmedabad",
  "cod": 200
}
```

Current Weather
Temperature 26.04
Pressure 1019
Humidity 18
Wind Speed 3.09
Visibility 6000

Air Pollution API

<https://openweathermap.org/api/air-pollution>

Used to obtain current air pollution data, air pollution forecast and also past air pollution data of any city

Current air pollution data

```
In [ ]: quality_indices = {1:'Good', 2:'Fair', 3:'Moderate', 4:'Poor', 5:'Very Poor'}
def display_current_air_pollution():
    api_key = input('Enter your API key: ')
    request_url = "http://api.openweathermap.org/data/2.5/air_pollution?lat=" + str(lat)
    response = requests.get(request_url)
    current_pollution = response.json()
    print(json.dumps(current_pollution, indent = 5))
    index = current_pollution['list'][0]['main']['aqi']
    print('Current Air Quality: ', quality_indices[index])

display_current_air_pollution()
```

Enter your API key: e9964ca2d303da57ca7c4b80c4afedf3

```
{
  "coord": {
    "lon": 72.5797,
    "lat": 23.0216
  },
  "list": [
    {
      "main": {
        "aqi": 3
      },
      "components": {
        "co": 343.8,
        "no": 2.32,
        "no2": 11.31,
        "o3": 130.18,
        "so2": 32.42,
        "pm2_5": 22.45,
        "pm10": 40,
        "nh3": 6.02
      },
      "dt": 1676272617
    }
  ]
}
Current Air Quality: Moderate
```

Exercise: Generate Weather Forecast for the next 5 Days in Ahmedabad:

<https://openweathermap.org/forecast5>

```
In [ ]: def five_day_forecast():
    api_key = input('Enter your API key: ')
    request_url = "http://api.openweathermap.org/data/2.5/forecast?lat=" + str(lat)
    response = requests.get(request_url)
    five_day = response.json()
    print(json.dumps(five_day, indent = 5))
    return five_day
```

```
In [ ]: five_day = five_day_forecast()
```


Enter your API key: 172dca3d92b0f6a90267417165f077dc

```
{
  "cod": "200",
  "message": 0,
  "cnt": 40,
  "list": [
    {
      "dt": 1676365200,
      "main": {
        "temp": 31.04,
        "feels_like": 28.98,
        "temp_min": 31.04,
        "temp_max": 33.02,
        "pressure": 1014,
        "sea_level": 1014,
        "grnd_level": 1008,
        "humidity": 15,
        "temp_kf": -1.98
      },
      "weather": [
        {
          "id": 800,
          "main": "Clear",
          "description": "clear sky",
          "icon": "01d"
        }
      ],
      "clouds": {
        "all": 0
      },
      "wind": {
        "speed": 1.59,
        "deg": 29,
        "gust": 1.67
      },
      "visibility": 10000,
      "pop": 0,
      "sys": {
        "pod": "d"
      },
      "dt_txt": "2023-02-14 09:00:00"
    },
    {
      "dt": 1676376000,
      "main": {
        "temp": 31.72,
        "feels_like": 29.52,
        "temp_min": 31.72,
        "temp_max": 33.08,
        "pressure": 1013,
        "sea_level": 1013,
        "grnd_level": 1006,
        "humidity": 12,
        "temp_kf": -1.36
      },
      "weather": [
        {
          "id": 800,
          "main": "Clear",
          "description": "clear sky",
          "icon": "01d"
        }
      ],
      "clouds": {
```

```

        "all": 0
    },
    "wind": {
        "speed": 0.84,
        "deg": 47,
        "gust": 1.5
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "d"
    },
    "dt_txt": "2023-02-14 12:00:00"
},
{
    "dt": 1676386800,
    "main": {
        "temp": 27.29,
        "feels_like": 26.1,
        "temp_min": 25.42,
        "temp_max": 27.29,
        "pressure": 1013,
        "sea_level": 1013,
        "grnd_level": 1008,
        "humidity": 12,
        "temp_kf": 1.87
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01n"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 1.84,
        "deg": 327,
        "gust": 2.79
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2023-02-14 15:00:00"
},
{
    "dt": 1676397600,
    "main": {
        "temp": 22.75,
        "feels_like": 21.45,
        "temp_min": 22.75,
        "temp_max": 22.75,
        "pressure": 1014,
        "sea_level": 1014,
        "grnd_level": 1008,
        "humidity": 14,
        "temp_kf": 0
    },
    "weather": [

```

```

        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01n"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 3.37,
        "deg": 342,
        "gust": 4.82
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2023-02-14 18:00:00"
},
{
    "dt": 1676408400,
    "main": {
        "temp": 20.94,
        "feels_like": 19.56,
        "temp_min": 20.94,
        "temp_max": 20.94,
        "pressure": 1013,
        "sea_level": 1013,
        "grnd_level": 1007,
        "humidity": 18,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01n"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 3.04,
        "deg": 352,
        "gust": 4.22
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2023-02-14 21:00:00"
},
{
    "dt": 1676419200,
    "main": {
        "temp": 19.93,
        "feels_like": 18.45,
        "temp_min": 19.93,

```

```

        "temp_max": 19.93,
        "pressure": 1012,
        "sea_level": 1012,
        "grnd_level": 1007,
        "humidity": 18,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01n"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 2.81,
        "deg": 8,
        "gust": 3.67
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2023-02-15 00:00:00"
},
{
    "dt": 1676430000,
    "main": {
        "temp": 22.07,
        "feels_like": 20.7,
        "temp_min": 22.07,
        "temp_max": 22.07,
        "pressure": 1015,
        "sea_level": 1015,
        "grnd_level": 1009,
        "humidity": 14,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01d"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 2.23,
        "deg": 43,
        "gust": 3.44
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "d"
    },

```

```
    "dt_txt": "2023-02-15 03:00:00"
  },
  {
    "dt": 1676440800,
    "main": {
      "temp": 31.15,
      "feels_like": 29.11,
      "temp_min": 31.15,
      "temp_max": 31.15,
      "pressure": 1015,
      "sea_level": 1015,
      "grnd_level": 1009,
      "humidity": 8,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 800,
        "main": "Clear",
        "description": "clear sky",
        "icon": "01d"
      }
    ],
    "clouds": {
      "all": 0
    },
    "wind": {
      "speed": 1.9,
      "deg": 60,
      "gust": 2.18
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
      "pod": "d"
    },
    "dt_txt": "2023-02-15 06:00:00"
  },
  {
    "dt": 1676451600,
    "main": {
      "temp": 35.62,
      "feels_like": 32.81,
      "temp_min": 35.62,
      "temp_max": 35.62,
      "pressure": 1011,
      "sea_level": 1011,
      "grnd_level": 1005,
      "humidity": 6,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 800,
        "main": "Clear",
        "description": "clear sky",
        "icon": "01d"
      }
    ],
    "clouds": {
      "all": 0
    },
    "wind": {
      "speed": 1.26,
```

```

        "deg": 6,
        "gust": 2.06
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "d"
    },
    "dt_txt": "2023-02-15 09:00:00"
},
{
    "dt": 1676462400,
    "main": {
        "temp": 35.68,
        "feels_like": 32.87,
        "temp_min": 35.68,
        "temp_max": 35.68,
        "pressure": 1010,
        "sea_level": 1010,
        "grnd_level": 1004,
        "humidity": 7,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01d"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 0.43,
        "deg": 233,
        "gust": 1.21
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "d"
    },
    "dt_txt": "2023-02-15 12:00:00"
},
{
    "dt": 1676473200,
    "main": {
        "temp": 27.25,
        "feels_like": 26.13,
        "temp_min": 27.25,
        "temp_max": 27.25,
        "pressure": 1012,
        "sea_level": 1012,
        "grnd_level": 1006,
        "humidity": 15,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",

```

```

        "icon": "01n"
    },
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 2.91,
        "deg": 304,
        "gust": 4.01
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2023-02-15 15:00:00"
},
{
    "dt": 1676484000,
    "main": {
        "temp": 24.37,
        "feels_like": 23.36,
        "temp_min": 24.37,
        "temp_max": 24.37,
        "pressure": 1013,
        "sea_level": 1013,
        "grnd_level": 1007,
        "humidity": 19,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01n"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 3.35,
        "deg": 345,
        "gust": 5.12
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2023-02-15 18:00:00"
},
{
    "dt": 1676494800,
    "main": {
        "temp": 23.32,
        "feels_like": 22.23,
        "temp_min": 23.32,
        "temp_max": 23.32,
        "pressure": 1012,
        "sea_level": 1012,
        "grnd_level": 1006,

```

```

        "humidity": 20,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01n"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 3.61,
        "deg": 346,
        "gust": 5.56
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2023-02-15 21:00:00"
},
{
    "dt": 1676505600,
    "main": {
        "temp": 21.61,
        "feels_like": 20.35,
        "temp_min": 21.61,
        "temp_max": 21.61,
        "pressure": 1012,
        "sea_level": 1012,
        "grnd_level": 1006,
        "humidity": 20,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01n"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 2.81,
        "deg": 11,
        "gust": 3.59
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2023-02-16 00:00:00"
},
{
    "dt": 1676516400,

```



```

"main": {
  "temp": 23.71,
  "feels_like": 22.58,
  "temp_min": 23.71,
  "temp_max": 23.71,
  "pressure": 1015,
  "sea_level": 1015,
  "grnd_level": 1009,
  "humidity": 17,
  "temp_kf": 0
},
"weather": [
  {
    "id": 800,
    "main": "Clear",
    "description": "clear sky",
    "icon": "01d"
  }
],
"clouds": {
  "all": 0
},
"wind": {
  "speed": 2.43,
  "deg": 39,
  "gust": 4.05
},
"visibility": 10000,
"pop": 0,
"sys": {
  "pod": "d"
},
"dt_txt": "2023-02-16 03:00:00"
},
{
  "dt": 1676527200,
  "main": {
    "temp": 33.02,
    "feels_like": 30.62,
    "temp_min": 33.02,
    "temp_max": 33.02,
    "pressure": 1015,
    "sea_level": 1015,
    "grnd_level": 1009,
    "humidity": 10,
    "temp_kf": 0
  },
  "weather": [
    {
      "id": 800,
      "main": "Clear",
      "description": "clear sky",
      "icon": "01d"
    }
  ],
  "clouds": {
    "all": 0
  },
  "wind": {
    "speed": 2.14,
    "deg": 60,
    "gust": 2.4
  },
  "visibility": 10000,

```

```

    "pop": 0,
    "sys": {
      "pod": "d"
    },
    "dt_txt": "2023-02-16 06:00:00"
  },
  {
    "dt": 1676538000,
    "main": {
      "temp": 37.4,
      "feels_like": 34.32,
      "temp_min": 37.4,
      "temp_max": 37.4,
      "pressure": 1011,
      "sea_level": 1011,
      "grnd_level": 1006,
      "humidity": 7,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 801,
        "main": "Clouds",
        "description": "few clouds",
        "icon": "02d"
      }
    ],
    "clouds": {
      "all": 21
    },
    "wind": {
      "speed": 1.45,
      "deg": 359,
      "gust": 2.58
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
      "pod": "d"
    },
    "dt_txt": "2023-02-16 09:00:00"
  },
  {
    "dt": 1676548800,
    "main": {
      "temp": 37.47,
      "feels_like": 34.33,
      "temp_min": 37.47,
      "temp_max": 37.47,
      "pressure": 1010,
      "sea_level": 1010,
      "grnd_level": 1005,
      "humidity": 6,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 801,
        "main": "Clouds",
        "description": "few clouds",
        "icon": "02d"
      }
    ],
    "clouds": {

```

```

        "all": 11
    },
    "wind": {
        "speed": 2.49,
        "deg": 296,
        "gust": 3.26
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "d"
    },
    "dt_txt": "2023-02-16 12:00:00"
},
{
    "dt": 1676559600,
    "main": {
        "temp": 28.03,
        "feels_like": 26.74,
        "temp_min": 28.03,
        "temp_max": 28.03,
        "pressure": 1013,
        "sea_level": 1013,
        "grnd_level": 1007,
        "humidity": 19,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01n"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 3.84,
        "deg": 311,
        "gust": 8.29
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2023-02-16 15:00:00"
},
{
    "dt": 1676570400,
    "main": {
        "temp": 25.32,
        "feels_like": 24.43,
        "temp_min": 25.32,
        "temp_max": 25.32,
        "pressure": 1014,
        "sea_level": 1014,
        "grnd_level": 1008,
        "humidity": 20,
        "temp_kf": 0
    },
    "weather": [

```

```

        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01n"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 2.87,
        "deg": 352,
        "gust": 4.51
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2023-02-16 18:00:00"
},
{
    "dt": 1676581200,
    "main": {
        "temp": 23.97,
        "feels_like": 22.94,
        "temp_min": 23.97,
        "temp_max": 23.97,
        "pressure": 1012,
        "sea_level": 1012,
        "grnd_level": 1007,
        "humidity": 20,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01n"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 3.59,
        "deg": 353,
        "gust": 5.91
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2023-02-16 21:00:00"
},
{
    "dt": 1676592000,
    "main": {
        "temp": 22.43,
        "feels_like": 21.25,
        "temp_min": 22.43,

```

```

        "temp_max": 22.43,
        "pressure": 1012,
        "sea_level": 1012,
        "grnd_level": 1007,
        "humidity": 20,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01n"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 2.63,
        "deg": 14,
        "gust": 3.2
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2023-02-17 00:00:00"
},
{
    "dt": 1676602800,
    "main": {
        "temp": 24.52,
        "feels_like": 23.52,
        "temp_min": 24.52,
        "temp_max": 24.52,
        "pressure": 1015,
        "sea_level": 1015,
        "grnd_level": 1009,
        "humidity": 19,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01d"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 2.19,
        "deg": 17,
        "gust": 3.29
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "d"
    },

```

```
    "dt_txt": "2023-02-17 03:00:00"
  },
  {
    "dt": 1676613600,
    "main": {
      "temp": 33.64,
      "feels_like": 31.15,
      "temp_min": 33.64,
      "temp_max": 33.64,
      "pressure": 1015,
      "sea_level": 1015,
      "grnd_level": 1010,
      "humidity": 10,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 800,
        "main": "Clear",
        "description": "clear sky",
        "icon": "01d"
      }
    ],
    "clouds": {
      "all": 0
    },
    "wind": {
      "speed": 2.33,
      "deg": 29,
      "gust": 2.82
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
      "pod": "d"
    },
    "dt_txt": "2023-02-17 06:00:00"
  },
  {
    "dt": 1676624400,
    "main": {
      "temp": 37.67,
      "feels_like": 34.5,
      "temp_min": 37.67,
      "temp_max": 37.67,
      "pressure": 1011,
      "sea_level": 1011,
      "grnd_level": 1006,
      "humidity": 6,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 800,
        "main": "Clear",
        "description": "clear sky",
        "icon": "01d"
      }
    ],
    "clouds": {
      "all": 0
    },
    "wind": {
      "speed": 1.04,
```

```

        "deg": 19,
        "gust": 2.85
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "d"
    },
    "dt_txt": "2023-02-17 09:00:00"
},
{
    "dt": 1676635200,
    "main": {
        "temp": 37.56,
        "feels_like": 34.41,
        "temp_min": 37.56,
        "temp_max": 37.56,
        "pressure": 1010,
        "sea_level": 1010,
        "grnd_level": 1005,
        "humidity": 6,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01d"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 1.99,
        "deg": 285,
        "gust": 2.44
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "d"
    },
    "dt_txt": "2023-02-17 12:00:00"
},
{
    "dt": 1676646000,
    "main": {
        "temp": 28.21,
        "feels_like": 26.8,
        "temp_min": 28.21,
        "temp_max": 28.21,
        "pressure": 1013,
        "sea_level": 1013,
        "grnd_level": 1007,
        "humidity": 15,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",

```

```
        "icon": "01n"
      }
    ],
    "clouds": {
      "all": 0
    },
    "wind": {
      "speed": 3.5,
      "deg": 301,
      "gust": 5.75
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
      "pod": "n"
    },
    "dt_txt": "2023-02-17 15:00:00"
  },
  {
    "dt": 1676656800,
    "main": {
      "temp": 25.68,
      "feels_like": 24.77,
      "temp_min": 25.68,
      "temp_max": 25.68,
      "pressure": 1013,
      "sea_level": 1013,
      "grnd_level": 1007,
      "humidity": 18,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 800,
        "main": "Clear",
        "description": "clear sky",
        "icon": "01n"
      }
    ],
    "clouds": {
      "all": 0
    },
    "wind": {
      "speed": 3.25,
      "deg": 344,
      "gust": 4.77
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
      "pod": "n"
    },
    "dt_txt": "2023-02-17 18:00:00"
  },
  {
    "dt": 1676667600,
    "main": {
      "temp": 24,
      "feels_like": 22.87,
      "temp_min": 24,
      "temp_max": 24,
      "pressure": 1012,
      "sea_level": 1012,
      "grnd_level": 1006,
```



```

        "humidity": 16,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01n"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 3.57,
        "deg": 359,
        "gust": 5.74
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2023-02-17 21:00:00"
},
{
    "dt": 1676678400,
    "main": {
        "temp": 22.1,
        "feels_like": 20.81,
        "temp_min": 22.1,
        "temp_max": 22.1,
        "pressure": 1012,
        "sea_level": 1012,
        "grnd_level": 1006,
        "humidity": 17,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01n"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 2.94,
        "deg": 28,
        "gust": 4.29
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2023-02-18 00:00:00"
},
{
    "dt": 1676689200,

```

```

"main": {
  "temp": 24.44,
  "feels_like": 23.44,
  "temp_min": 24.44,
  "temp_max": 24.44,
  "pressure": 1014,
  "sea_level": 1014,
  "grnd_level": 1008,
  "humidity": 19,
  "temp_kf": 0
},
"weather": [
  {
    "id": 800,
    "main": "Clear",
    "description": "clear sky",
    "icon": "01d"
  }
],
"clouds": {
  "all": 0
},
"wind": {
  "speed": 2.76,
  "deg": 69,
  "gust": 4.98
},
"visibility": 10000,
"pop": 0,
"sys": {
  "pod": "d"
},
"dt_txt": "2023-02-18 03:00:00"
},
{
  "dt": 1676700000,
  "main": {
    "temp": 34.04,
    "feels_like": 31.49,
    "temp_min": 34.04,
    "temp_max": 34.04,
    "pressure": 1014,
    "sea_level": 1014,
    "grnd_level": 1009,
    "humidity": 10,
    "temp_kf": 0
  },
  "weather": [
    {
      "id": 800,
      "main": "Clear",
      "description": "clear sky",
      "icon": "01d"
    }
  ],
  "clouds": {
    "all": 0
  },
  "wind": {
    "speed": 3.46,
    "deg": 85,
    "gust": 4.09
  },
  "visibility": 10000,

```

```

    "pop": 0,
    "sys": {
      "pod": "d"
    },
    "dt_txt": "2023-02-18 06:00:00"
  },
  {
    "dt": 1676710800,
    "main": {
      "temp": 38.09,
      "feels_like": 34.9,
      "temp_min": 38.09,
      "temp_max": 38.09,
      "pressure": 1010,
      "sea_level": 1010,
      "grnd_level": 1004,
      "humidity": 7,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 800,
        "main": "Clear",
        "description": "clear sky",
        "icon": "01d"
      }
    ],
    "clouds": {
      "all": 0
    },
    "wind": {
      "speed": 2.28,
      "deg": 52,
      "gust": 2.26
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
      "pod": "d"
    },
    "dt_txt": "2023-02-18 09:00:00"
  },
  {
    "dt": 1676721600,
    "main": {
      "temp": 37.6,
      "feels_like": 34.44,
      "temp_min": 37.6,
      "temp_max": 37.6,
      "pressure": 1009,
      "sea_level": 1009,
      "grnd_level": 1004,
      "humidity": 6,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 800,
        "main": "Clear",
        "description": "clear sky",
        "icon": "01d"
      }
    ],
    "clouds": {

```

```

        "all": 0
    },
    "wind": {
        "speed": 1.41,
        "deg": 25,
        "gust": 1.8
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "d"
    },
    "dt_txt": "2023-02-18 12:00:00"
},
{
    "dt": 1676732400,
    "main": {
        "temp": 29.38,
        "feels_like": 27.65,
        "temp_min": 29.38,
        "temp_max": 29.38,
        "pressure": 1011,
        "sea_level": 1011,
        "grnd_level": 1006,
        "humidity": 11,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01n"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 2.14,
        "deg": 335,
        "gust": 3.9
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2023-02-18 15:00:00"
},
{
    "dt": 1676743200,
    "main": {
        "temp": 25.84,
        "feels_like": 24.85,
        "temp_min": 25.84,
        "temp_max": 25.84,
        "pressure": 1012,
        "sea_level": 1012,
        "grnd_level": 1007,
        "humidity": 14,
        "temp_kf": 0
    },
    "weather": [

```

```

        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01n"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 2.37,
        "deg": 342,
        "gust": 3.64
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2023-02-18 18:00:00"
},
{
    "dt": 1676754000,
    "main": {
        "temp": 24.21,
        "feels_like": 23.16,
        "temp_min": 24.21,
        "temp_max": 24.21,
        "pressure": 1011,
        "sea_level": 1011,
        "grnd_level": 1005,
        "humidity": 18,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01n"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 2.55,
        "deg": 345,
        "gust": 3.08
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2023-02-18 21:00:00"
},
{
    "dt": 1676764800,
    "main": {
        "temp": 22.61,
        "feels_like": 21.4,
        "temp_min": 22.61,

```

```

        "temp_max": 22.61,
        "pressure": 1011,
        "sea_level": 1011,
        "grnd_level": 1005,
        "humidity": 18,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01n"
        }
    ],
    "clouds": {
        "all": 1
    },
    "wind": {
        "speed": 2.51,
        "deg": 16,
        "gust": 3.07
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2023-02-19 00:00:00"
},
{
    "dt": 1676775600,
    "main": {
        "temp": 24.86,
        "feels_like": 23.77,
        "temp_min": 24.86,
        "temp_max": 24.86,
        "pressure": 1013,
        "sea_level": 1013,
        "grnd_level": 1007,
        "humidity": 14,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 800,
            "main": "Clear",
            "description": "clear sky",
            "icon": "01d"
        }
    ],
    "clouds": {
        "all": 0
    },
    "wind": {
        "speed": 1.79,
        "deg": 43,
        "gust": 2.83
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "d"
    },

```

```

        "dt_txt": "2023-02-19 03:00:00"
    },
    {
        "dt": 1676786400,
        "main": {
            "temp": 33.9,
            "feels_like": 31.37,
            "temp_min": 33.9,
            "temp_max": 33.9,
            "pressure": 1013,
            "sea_level": 1013,
            "grnd_level": 1007,
            "humidity": 9,
            "temp_kf": 0
        },
        "weather": [
            {
                "id": 800,
                "main": "Clear",
                "description": "clear sky",
                "icon": "01d"
            }
        ],
        "clouds": {
            "all": 0
        },
        "wind": {
            "speed": 1.35,
            "deg": 78,
            "gust": 1.31
        },
        "visibility": 10000,
        "pop": 0,
        "sys": {
            "pod": "d"
        },
        "dt_txt": "2023-02-19 06:00:00"
    }
],
"city": {
    "id": 1279233,
    "name": "Ahmedabad",
    "coord": {
        "lat": 23.0216,
        "lon": 72.5797
    },
    "country": "IN",
    "population": 3719710,
    "timezone": 19800,
    "sunrise": 1676338988,
    "sunset": 1676379891
}
}

```

```
In [ ]: D = {"date_time":[], "temp":[], "pressure":[], "humidity":[], "weather":[]}
```

```

for i in five_day['list']:
    D["date_time"].append(i["dt_txt"])
    D["temp"].append(i["main"]["temp"])
    D["pressure"].append(i["main"]["temp"])
    D["humidity"].append(i["main"]["humidity"])
    D["weather"].append(i["weather"][0]["description"])

```

```
In [ ]: import pandas as pd  
df = pd.DataFrame(D)  
df.head(40)
```


Out[]:

	date_time	temp	pressure	humidity	weather
0	2023-02-14 09:00:00	31.04	31.04	15	clear sky
1	2023-02-14 12:00:00	31.72	31.72	12	clear sky
2	2023-02-14 15:00:00	27.29	27.29	12	clear sky
3	2023-02-14 18:00:00	22.75	22.75	14	clear sky
4	2023-02-14 21:00:00	20.94	20.94	18	clear sky
5	2023-02-15 00:00:00	19.93	19.93	18	clear sky
6	2023-02-15 03:00:00	22.07	22.07	14	clear sky
7	2023-02-15 06:00:00	31.15	31.15	8	clear sky
8	2023-02-15 09:00:00	35.62	35.62	6	clear sky
9	2023-02-15 12:00:00	35.68	35.68	7	clear sky
10	2023-02-15 15:00:00	27.25	27.25	15	clear sky
11	2023-02-15 18:00:00	24.37	24.37	19	clear sky
12	2023-02-15 21:00:00	23.32	23.32	20	clear sky
13	2023-02-16 00:00:00	21.61	21.61	20	clear sky
14	2023-02-16 03:00:00	23.71	23.71	17	clear sky
15	2023-02-16 06:00:00	33.02	33.02	10	clear sky
16	2023-02-16 09:00:00	37.40	37.40	7	few clouds
17	2023-02-16 12:00:00	37.47	37.47	6	few clouds
18	2023-02-16 15:00:00	28.03	28.03	19	clear sky
19	2023-02-16 18:00:00	25.32	25.32	20	clear sky
20	2023-02-16 21:00:00	23.97	23.97	20	clear sky
21	2023-02-17 00:00:00	22.43	22.43	20	clear sky
22	2023-02-17 03:00:00	24.52	24.52	19	clear sky
23	2023-02-17 06:00:00	33.64	33.64	10	clear sky
24	2023-02-17 09:00:00	37.67	37.67	6	clear sky
25	2023-02-17 12:00:00	37.56	37.56	6	clear sky
26	2023-02-17 15:00:00	28.21	28.21	15	clear sky
27	2023-02-17 18:00:00	25.68	25.68	18	clear sky
28	2023-02-17 21:00:00	24.00	24.00	16	clear sky
29	2023-02-18 00:00:00	22.10	22.10	17	clear sky
30	2023-02-18 03:00:00	24.44	24.44	19	clear sky
31	2023-02-18 06:00:00	34.04	34.04	10	clear sky
32	2023-02-18 09:00:00	38.09	38.09	7	clear sky
33	2023-02-18 12:00:00	37.60	37.60	6	clear sky
34	2023-02-18 15:00:00	29.38	29.38	11	clear sky
35	2023-02-18 18:00:00	25.84	25.84	14	clear sky

	date_time	temp	pressure	humidity	weather
36	2023-02-18 21:00:00	24.21	24.21	18	clear sky
37	2023-02-19 00:00:00	22.61	22.61	18	clear sky
38	2023-02-19 03:00:00	24.86	24.86	14	clear sky
39	2023-02-19 06:00:00	33.90	33.90	9	clear sky

Similarly, air pollution forecast and history can be obtained using the air pollution API